KALAM (களம்) — AGRICULTURAL INSIGHTS ATLAS

Ambasamudram & Cheranmahadevi Taluks, Tirunelveli · Multi-Year EO Analytics (2023–2026)

CROPPING INTENSITY

4.8% Double-Cropped Cropland

AUDIT READINESS

205 Targeted Audit Parcels (GPS)

ANNUAL WATER DEMAND

37.71 Million m³ (MCM) Canal Requirement

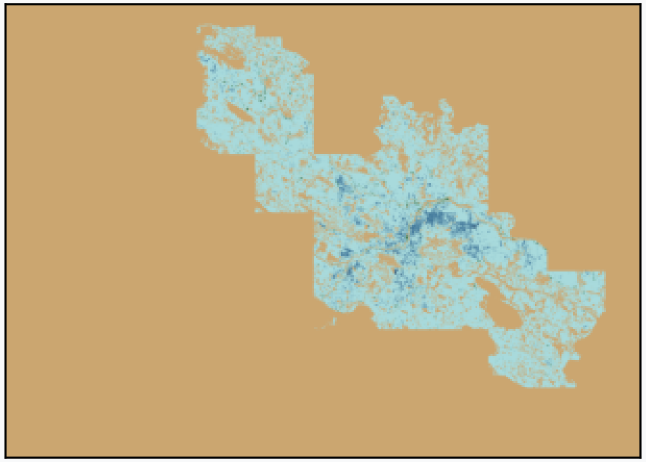
EXECUTIVE INSIGHT SUMMARY & POLICY ACTION ITEMS

- **Cropping Intensity:** 4.8% of cropland (1694.2 ha) in Ambasamudram & Cheranmahadevi is double-cropped with dual Kar and Pishanam cycles, alongside 1.1% under perennial banana canopy.
- **Transplanting Calendar:** Peak Kar transplanting occurs in Jun (31.9% of fields), while main Pishanam sowing peaks in Nov. 0.0% of Pishanam acreage experiences delayed transplanting beyond November.
- **Field Verification Priority:** 25.0% of parcels (511 fields) are flagged for targeted ground audit due to lower phenological confidence or inter-annual crop instability, with top-50 priority targets packaged for GPS navigation.
- **Village-Level Statistics:** Kallidaikurichi leads overall agricultural cultivation across both crops with 614.4 ha of paddy (68% share) and 7.7 ha of banana.
- **Irrigation Dependency & Water Access:** 46.6% of paddy acreage is concentrated within 500 m of a river channel, canal or irrigation tank. Banana accounts for 1.0% of cropland in the immediate 0–250 m corridor, decaying sharply beyond 1 km.
- **Crop Change & Farmland Dynamics:** Evaluated 4,825.0 ha of 4,825.0 ha cropland (confidence ≥ 0.60): 26.7% of paddy and 0.8% of banana parcels remained stable across 2023–2026, with 82.2 ha (1.7%) identified as candidate paddy-to-banana diversification.
- **Crop Health & NDVI Anomaly:** 74.6% of analyzed cropland displays normal or above-normal vegetative vigour. Localized moisture or transplanting delay stress affects 38.2% (1104.0 ha), predominantly along tail-end distributaries.
- **Indicative Gross Irrigation Requirement:** Total indicative gross irrigation water requirement is 37.7 MCM/year. Dual-season paddy accounts for 73.3% (27.6 MCM) of total demand, while perennial banana accounts for 2.2% (0.8 MCM).
- **Segment Size Distribution (Image-Derived):** 19.8% of image-derived field segments (9.9% of agricultural area) are below 1.0 ha. The mean segment size is 1.64 ha with a median of 1.41 ha.

Kalam © 2026 · Problem Statement 1 · SustainAI / Mapathon · Contains modified Copernicus Sentinel data

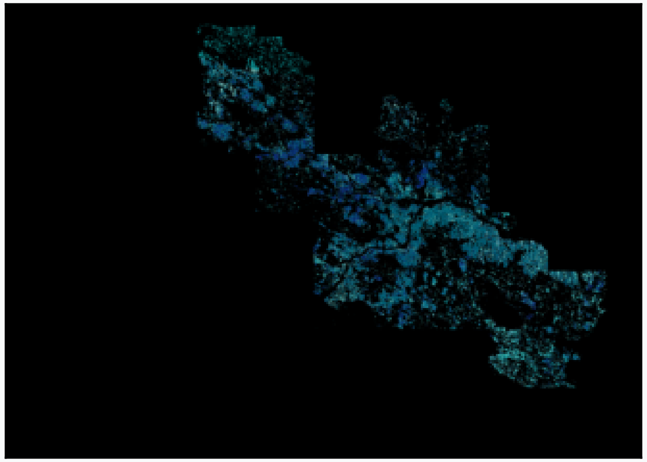
PHENOLOGY DYNAMICS & VERIFICATION PRIORITIZATION

01. Cropping Intensity



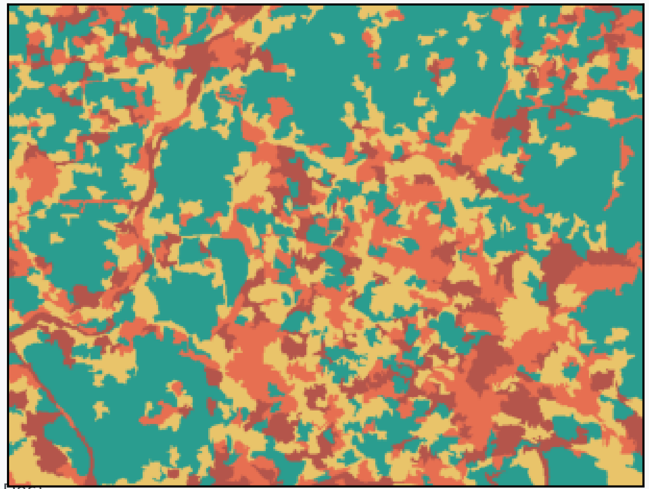
ted (175729.3 ha, 499.7%) | Single Crop (33870.7 ha, 96.3%) | Double Crop (1694.2 ha, 4.8%) | Triple Crop (31.8 ha, 0.1%)

02. Transplanting Calendar



Kar Peak (Jun–Jul) | Kar Late (Aug) | Pishanam Peak (Oct) | Pishanam Delayed (Nov–Dec)

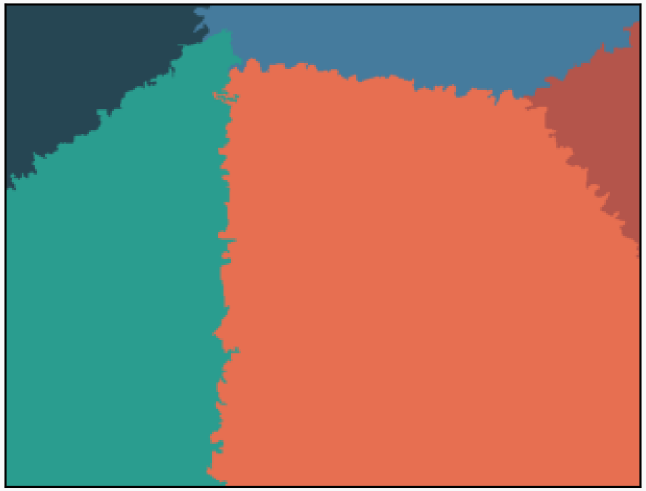
03. Field Audit Priority



Critical Priority (Top 10%) | High Priority (Next 15%) | Medium Priority (Next 25%) | Low Priority (Baseline 50

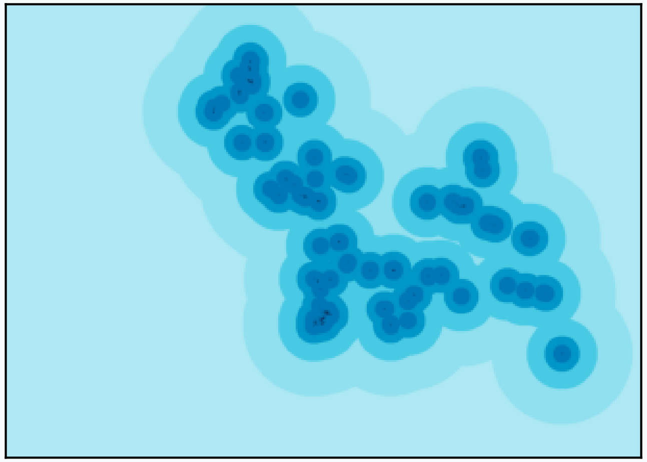
CANAL BUFFER DEPENDENCY & MULTI-YEAR CROP TRANSITIONS

04. Village Crop Allocations



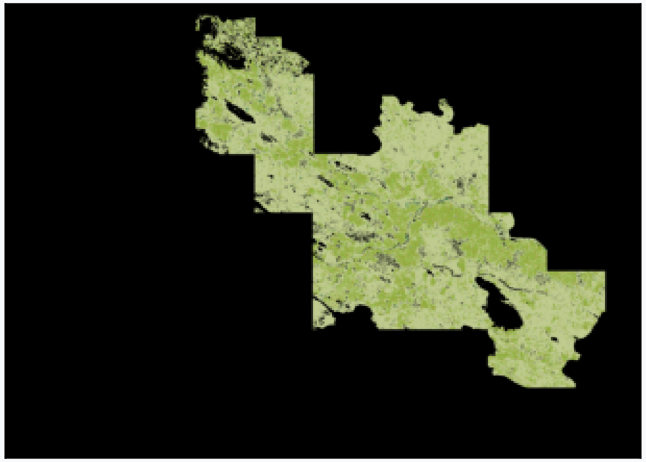
Urban (Paddy intensive) | Kallidaikurichi (Paddy intensive) | Veeravanallur (Diversified / Fallow) | Mukkudal (Paddy intensive)

05. Canal Buffer Decay



0 – 250 m | 250 – 500 m | 500 – 1,000 m | 1,000 – 2,000 m

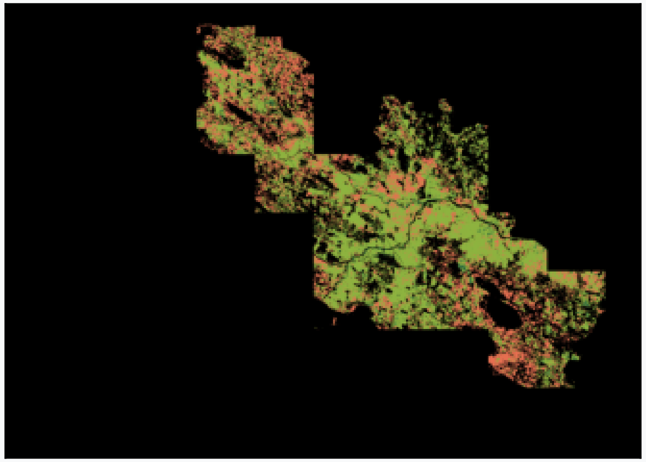
06. Crop Transitions (23–26)



Stable Paddy | Stable Banana | Candidate Paddy → Banana | Banana → Paddy

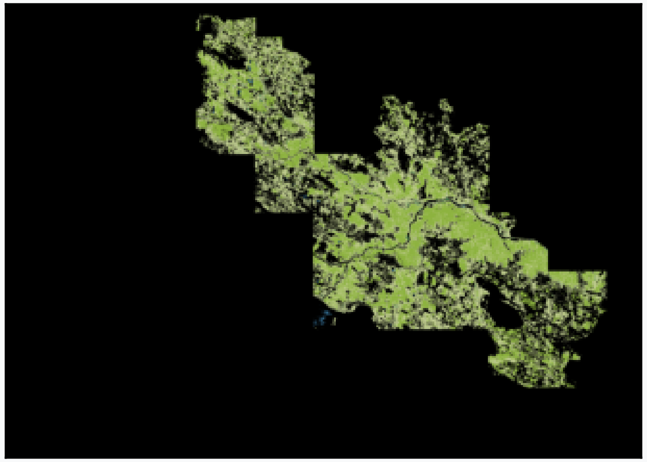
CROP VIGOUR, WATER DEMAND & PARCEL MORPHOLOGY

07. NDVI Vigour Anomaly



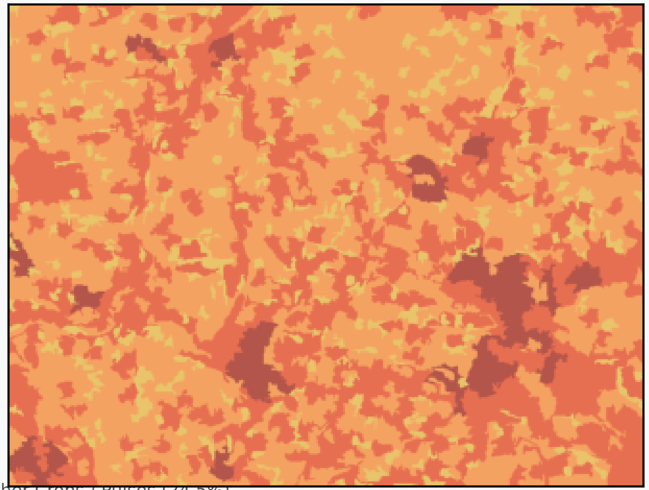
s (25.0 ha, 0.9%) | Mild Stress (1079.0 ha, 37.3%) | Normal Vigour (2032.3 ha, 70.3%) | Above Normal (123.9 ha, 4.3%)

08. Water Demand Budget



Paddy (Kar Season) (28.9%) | Paddy (Pisanam Season) (44.4%) | Banana (Annual Perennial) (2.2%) | Other Crops / Pulses (24.5%)

09. Smallholder Fragmentation



Marginal (< 0.5 ha) (0.0% area, 0.0% count) | Small (0.5 – 1.0 ha) (9.9% area, 19.8% count) | Semi-Medium (1.0 – 2.0 ha) (50.6% area, 58.1% count) | M